A Thirteen-Variable Family under the Rational *abc* Conjecture

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Abstract

Assuming the rational *abc* conjecture, together with the specified elliptic group, pole-realization and simultaneous prime-pattern inputs, we construct a family of integer polynomials in thirteen total variables. Every sufficiently large fixed odd input shift gives rational input projection exactly $\mathbb{Z}$. Each member has total degree less than 2^{44}. A logarithmic height estimate for odd contact radicals and a height estimate for a sum of sixth powers allow the rational input and one elliptic index to be bounded together. Only three elliptic points and two pole certificates occur. All six local tests have order one and use the elliptic formal parameter itself. The coefficient threshold is ineffective. The additional *abc* hypothesis is essential to the proof given here; this is not a reduction under the arithmetic premises of the separate twenty-seven-variable theorem alone.

1 Statement and hypotheses

All variable counts include the input. For a rational number a/b in lowest terms with $b > 0$, write $h(a/b) = \log \max(|a|, b)$. The rational *abc* conjecture means that for every $\epsilon > 0$ there exists $K_\epsilon \geq 1$ such that

$$c \leq K_\epsilon \operatorname{rad}(abc)^{1+\epsilon} \quad (a + b = c,\ a, b, c \text{ positive and pairwise coprime}).$$

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad D = (3 - \sqrt{2})^2, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

In addition to *abc*, retain the following arithmetic inputs with exactly their stated scope in [3, 2].

1. $2E(K) = \mathbb{Z}P$, with the indicated non-torsion generator and the source index-height bound. The direct point chart below therefore describes the nonzero multiples of P.

2. The pole system, for $b \neq 0$ and $B_0 = 17b$,

$$p^2 = B_0(1 - e)(1 + 3e), \qquad B_0 q^2 = (s^2 - B_0 e(B_0 - b^2))(s^2 + 3B_0 e(B_0 - b^2)),$$

excludes negative odd valuations of b outside $\{2, 3, 17\}$. For a prepared input

$$\begin{gathered} 0 < b < 17, \quad b \text{ has square reduced denominator}, \quad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \\ v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2, \end{gathered} \tag{1}$$

the retained restricted-converse and descent hypotheses supply a witness with $pqs \neq 0$. The nonzero-witness consequence and its single conic chart are proved in [2, Lemma 2].

3. The simultaneous prime-pattern input supplies the specified prime contact numerators, Gaussian congruences, conjugate reduction orders, fixed divisibility and avoidance, and compatible real approximation for any fixed finite set of positive rational square scales. Its precise quantified version and fixed-avoidance deduction from Sun's theorem are given in [3]. The order of choices is respected in Section 6.

The exponent-nineteen contact-height input of the earlier source is not needed here. It is replaced by Lemma 2.1, which uses the additional *abc* assumption.

The proved one-point case of the aggregate trace-height theorem [5] gives constants $A > 0$, $J \geq 0$, depending only on E, P, such that

$$n^2 \leq Ah(T(n)) + J, \qquad T(n) = \mathrm{Tr}_{K/\mathbb{Q}}\, x([n]P) \quad (n \in \mathbb{Z} \setminus \{0\}). \tag{2}$$

This applies von Buhren's theorem after checking the absence of constant abelian translates for the fixed trace function. Its additive constant is ineffective. It is a proved height theorem, not an additional conjecture. For a nonzero multiple of P, the abscissa is a positive square in both real embeddings; the curve equation then gives $x > 3 \mp \sqrt{2}$ in the respective embeddings, hence $T(n) > 6$.

Theorem 1.1. *Under the assumptions above, there is a computable family*

$$F_B \in \mathbb{Z}[X, Y_1, \ldots, Y_{12}], \qquad B \geq 3 \text{ odd},$$

with the following properties.

1. *Every member is given by a complete finite integer coefficient specification and satisfies*

$$\deg F_B \leq 13155996618240 < 2^{44}.$$

2. *There exists a constant B_* such that every odd $B \geq B_*$ has rational input projection exactly $\mathbb{Z}$.*

3. *No numerical sufficient value of B_* is determined. The supplied member $B = 3$ is a coefficient-verification fixture, not a polynomial certified to define $\mathbb{Z}$.*

The integer B is a fixed coefficient parameter and is never a quantified coordinate.

2 Two consequences of *abc*

Put

$$\Theta = \left\{0, -\frac{1}{2}, -1, \frac{1}{2399}, -\frac{32}{407}\right\}, \qquad S_0 = \{p : p \leq 76176\}.$$

For $t \in \mathbb{Q} \setminus \Theta$, let $M_0(t)$ be the product of primes outside S_0 for which $v_p(t - \theta)$ is positive and odd for some $\theta \in \Theta$.

Lemma 2.1. *Assume rational abc. There is a constant $C_0 \geq 0$ such that*

$$h(t) \leq \frac{10}{9} \log M_0(t) + C_0 \qquad (t \in \mathbb{Q} \setminus \Theta).$$

Proof. For a primitive pair (u, v), consider the squarefree homogeneous binary form

$$G(u, v) = vu(u - v)(u - 2401v)(32u + 343v)$$

of degree five. Granville's consequence of *abc* [1, Proposition 1], with $\epsilon = 1/20$, gives

$$\log \operatorname{rad}(G(u,v)) \geq \frac{59}{20} \log \max(|u|,|v|) - C.$$

For a nonzero integer N, if $\operatorname{sf}(N)$ denotes the product of primes having odd valuation in N, then $\operatorname{rad}(N)^2 \leq |N| \operatorname{sf}(N)$, prime by prime. Outside S_0, two of the five primitive factors of G cannot vanish simultaneously. Thus its squarefree part outside S_0 is precisely the odd contact radical for $\infty, 0, 1, 2401, -343/32$. The omitted primes contribute a fixed bounded factor. Also $|G(u,v)| \leq C' \max(|u|,|v|)^5$. Combining the two estimates gives

$$\log M_5(u/v) \geq \frac{9}{10} h(u/v) - C''.$$

The integral linear fractional transformation $t \mapsto 2 + 1/t$ has determinant -1, sends the contacts Θ to the five original contacts, and changes logarithmic height by at most $\log 3$. Its contact radicals coincide outside S_0. These assertions follow directly by substituting a primitive numerator-denominator pair; their fixed exceptional determinants have prime divisors in S_0. Absorbing the fixed constants proves the claim. □

Lemma 2.2. *Assume rational abc. There is a constant $C_1 \geq 0$ such that*

$$h(u) + h(v) \leq \frac{4}{5} h(u^6 + v^6) + C_1 \qquad (u, v \in \mathbb{Q}).$$

One may take $C_1 = \frac{7}{15} \log 2 + \frac{11}{15} \log K_{1/11}$. For $(u,v) \neq (0,0)$ and $a = (u^6+v^6)^2$, equivalently,

$$h(u) + h(v) \leq \frac{2}{5} h(a) + C_1.$$

Proof. The assertion is immediate if a coordinate is zero, since $h(w^6) = 6h(w)$. Otherwise choose integers A', B', D_0, with $D_0 > 0$ and $\gcd(D_0, A', B') = 1$, representing $u = A'/D_0$, $v = B'/D_0$. Write $g = \gcd(|A'|, |B'|)$ and $A_0 = A'/g$, $B_0 = B'/g$. Then $\gcd(A_0, B_0) = \gcd(g, D_0) = 1$. Put $H = \max(|A_0|, |B_0|)$ and $N = A_0^6 + B_0^6$, so $H^6 \leq N \leq 2H^6$.

Let $k = \gcd(N, D_0^6)$, $m_0 = N/k$, and $n = D_0^6/k$. The fraction $s = u^6 + v^6 = m/n$, where $m = g^6 m_0$, is reduced. Put $M = \max(m,n)$, so $h(s) = \log M$. Every prime dividing N either divides D_0 or survives in m_0, giving $\operatorname{rad}(N) \leq D_0 m_0$. Apply *abc* at $\epsilon = 1/11$ to the pairwise coprime triple A_0^6, B_0^6, N:

$$H^{7/2} \leq K_{1/11}^{11/12} D_0 m_0.$$

Since $D_0^6 = nN/m_0 \leq 2nH^6/m_0$, this implies

$$H^{5/2} \leq 2^{1/6} K_{1/11}^{11/12} m_0^{5/6} n^{1/6}.$$

Moreover

$$\max(gH, D_0) \leq 2^{1/6} H (M/m_0)^{1/6}.$$

Taking logarithms and combining these inequalities gives

$$\begin{aligned} \log \max(gH, D_0) &\leq \frac{7}{30} \log 2 + \frac{11}{30} \log K_{1/11} + \frac{1}{6} \log m_0 + \frac{1}{15} \log n + \frac{1}{6} \log M \\ &\leq \frac{2}{5} \log M + \frac{1}{2} C_1. \end{aligned}$$

Each of $h(u), h(v)$ is at most the left-hand side. Adding proves the first assertion; $h(s^2) = 2h(s)$ proves the second. The common numerator factor and all denominator cancellations are thus included. □

3 The complete rational source

Fix an odd integer $B \geq 3$ and set

$$\beta = 2X^2 + B. \tag{3}$$

For rational $X = u/v$ in lowest terms, integrality of β implies $v^2 \mid 2$, hence $v = 1$. Thus $\beta \in \mathbb{Z}$ if and only if $X \in \mathbb{Z}$. For integer input β is odd, and $\beta \geq B$ for every rational input.

Retain three pairs of rational coordinates, one for each of the points A, H, U. For any such pair (h_0, h_1) put

$$h = h_0 + h_1\sqrt{2}, \quad x = h^2, \quad j^2 = h^4 - D, \quad y = hj.$$

The direct point-chart theorem of [2] proves that these are exactly the nonzero multiples of P. Each finite choice j is encoded by one scalar monic quartic as follows. Write $h^4 - D = a + b\sqrt{2}$, put $c = 2a + b$, $d = -a - b$, and write $(c + d\sqrt{2})^2(a + b\sqrt{2}) = a' + b'\sqrt{2}$. Then

$$Y^4 - 4a'Y^2 + 8(b')^2 = 0, \quad j = \frac{Y/2 + (b'/Y)\sqrt{2}}{c + d\sqrt{2}}.$$

Its discriminant is $2^{15}(b')^2(2a^2 - b^2)^4(a^2 - 2b^2)^2$, nonzero at every rational pair (h_0, h_1). Indeed $h^4 - D \neq 0$ by its norm, the two quadratic norm factors are anisotropic over $\mathbb{Q}$, and $b' = -4a^3 + 6ab^2 + 3b^3$ is anisotropic by Eisenstein at 3. The square-root chart is complete because its transformed rational component is $2(u^3 - 2v^3)$ for $j = u + v\sqrt{2}$, which is nonzero for $j \neq 0$. These are algebraic facts independent of *abc*.

Compute $V_0 = H - P$ and $V_1 = U - A$ by the usual affine addition formulas. Put $f = 1/x$, $z = -x/y$, let σ be conjugation of $K/\mathbb{Q}$, and define

$$L = \frac{z(U)}{z(H)}, \qquad L' = \sigma\left(\frac{z(V_1)}{z(V_0)}\right).$$

Use precisely the six field-valued tests

$$D_1 = f(H),\ D_2 = f(U),\ D_3 = \sigma f(V_0),\ D_4 = \sigma f(V_1),\ D_5 = L - L',\ D_6 = L - \beta. \tag{4}$$

Every test is assigned order one. No scalar-height pole or Lucas relation is present.

Let $T = \operatorname{Tr} x(A)$, and use the two positive rational square scales

$$a_0 = 1, \qquad a_1 = (\beta^6 + T^6)^2.$$

Retain one rational slope λ. For $r = 0, 1$, set

$$t_r = a_r\lambda, \qquad \delta_r = t_r - 1, \qquad F_r = \prod_{\theta \in \Theta}(t_r - \theta), \qquad R_r = \frac{F_r}{a_r^5 \delta_r^6}, \tag{5}$$

$$W_r = \prod_{j=1}^{6}\left(1 + \frac{D_j^2}{\delta_r^4 R_r}\right), \qquad b_r = \frac{R_r}{\operatorname{Tr}(W_r^{-1})^2}. \tag{6}$$

All displayed divisions, affine-chart denominators, F_r, δ_r, $\operatorname{Norm}(W_r)$, and $\operatorname{Tr}(W_r^{-1})$ are guarded nonzero. These conditions are part of the source.

For each $b = b_r$, retain independent rational parameters τ_r, s_r. Put $B_0 = 17b$ and

$$e = \frac{\tau_r^2 - B_0}{\tau_r^2 + 3B_0}, \qquad q^2 = \frac{(s_r^2 - B_0 e(B_0 - b^2))(s_r^2 + 3B_0 e(B_0 - b^2))}{B_0}.$$

Guard the denominator and require $q \neq 0$. This reconstructs the first pole ordinate as $p = -4B_0\tau_r/(\tau_r^2 + 3B_0)$. Each rational-square equation $q^2 = N_0/D_0$ is made monic by $Z = D_0 q$, giving $Z^2 = N_0 D_0$ with $Z \neq 0$ and $D_0 \neq 0$. No new unrestricted rational parameter is used for these roots.

4 The local implication

Lemma 4.1. *For any rational solution of the guarded source, write $A = [n]P$, $H = [h]P$, $U = [u]P$. If $\beta \neq n$ and N is the positive absolute numerator of $\beta - n$ in lowest terms, then*

$$M_0(t_r) \mid N \qquad (r = 0, 1).$$

Proof. Fix a prime $p \mid M_0(t_r)$ and abbreviate $R = R_r$, $W = W_r$. Outside S_0 only one contact factor has positive valuation, δ_r is a unit, and $v_p(R)$ is odd because a_r is a square. For each place $\mathfrak{p}$ above p, the two terms of a factor of W have valuations of different parity. Hence

$$v_{\mathfrak{p}}(W) = \sum_j \min\{0, 2v_{\mathfrak{p}}(D_j) - v_p(R)\} \leq 0.$$

Put $w = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \leq 0$. Then $v_p \operatorname{Tr}(W^{-1}) \geq -w$. The valuation of $b = R/\operatorname{Tr}(W^{-1})^2$ is odd, so the pole obstruction gives

$$0 < v_p(b) \leq v_p(R) + 2w.$$

Thus $\nu = v_p(R) > 0$ and $w > -\nu/2$. At a place attaining w, each summand in the valuation of W is at least w. It follows that $v_{\mathfrak{p}}(D_j) > \nu/4 > 0$, so all six tests have valuation at least one at that place.

The first four tests put H, U into the formal group at $\mathfrak{p}$ and V_0, V_1 into it at the conjugate place. At these good unramified places the formal logarithm satisfies

$$\log_E(Q) = z(Q)(1 + O(p^4)).$$

Indeed its first omitted term is a fixed multiple of $Df(Q)^2$; $v_p(f(Q)) \geq 2$ in the formal group. The remaining coefficient estimates, uniformly for $p > 76176$, are those proved in the formal-logarithm paper [6]. The exact logarithmic ratios are u/h and $(u - n)/(h - 1)$, respectively.

The affine guards imply $h \neq 0, 1$. At least one of $h, h-1$ is a p-adic unit, so the corresponding exact ratio is integral. Relative precision four makes its truncated ratio integral. The test $L - L' \equiv 0 \pmod{p}$ makes the other truncated ratio integral as well; the same relative precision then makes the other exact ratio integral. Relative errors are now absolute errors of valuation at least four. Consequently

$$L \equiv u/h, \qquad L' \equiv (u - n)/(h - 1), \qquad L \equiv L' \pmod{p}.$$

Multiplying by the integral numbers $h, h-1$ and subtracting yields $L \equiv n \pmod{p}$. The target test gives $L \equiv \beta \pmod{p}$. In particular β is p-integral, and p divides the reduced numerator of $\beta - n$. This holds at every consumed prime and proves the divisibility. The same argument includes inert primes; if their formal-group requirements are incompatible, no source solution occurs at that prime. □

5 Soundness and the uniform input shift

Suppose a rational source solution has $\beta \neq n$. By Lemma 4.1 and Lemma 2.1,

$$h(\lambda), h(a_1\lambda) \leq \frac{10}{9} \log N + C_0.$$

The quotient-height inequality therefore gives

$$h(a_1) \leq \frac{20}{9} \log N + 2C_0.$$

Lemma 2.2, with $u = \beta, v = T(n)$, implies

$$h(\beta) + h(T(n)) \le \frac{8}{9}\log N + C, \qquad C = \frac{4}{5}C_0 + C_1. \tag{7}$$

All constants here depend only on the fixed contact set and the fixed sixth-power map. They are independent of the shift B, the rational input, and every witness.

Writing $\beta = a/b$ in lowest terms, one has $N = |a - nb| \le e^{h(\beta)}(1 + |n|)$. Substitution in (7) gives

$$\frac{1}{9}h(\beta) + h(T(n)) \le \frac{8}{9}\log(1 + |n|) + C. \tag{8}$$

With (2), this implies

$$|n|^2 \le \frac{8A}{9}\log(1 + |n|) + AC + J \le \frac{8A}{9}|n| + AC + J.$$

Choose an integer $M > 8A/9 + \sqrt{AC + J}$. Then $|n| < M$, and (8) gives

$$h(\beta) \le 8\log(1 + M) + 9C =: H_*.$$

Choose $B_* > e^{H_*}$. For every fixed odd $B \ge B_*$, (3) gives $h(\beta) \ge \log\beta \ge \log B > H_*$, a contradiction. Thus $\beta = n \in \mathbb{Z}$, and the elementary input-map argument yields $X \in \mathbb{Z}$. This proves soundness uniformly for every sufficiently large shift. It does not determine that shift effectively.

6 Completeness

Fix an integer input X and an odd coefficient $B \ge 3$. Then $n = \beta$ is an odd integer at least three. Fix $A = [n]P$ and its two positive square scales before choosing any prime. They are distinct because $a_1 > 1$. Temporarily put $H = [h]P$ and $U = [hn]P$.

Choose a nonexceptional $h_* \in 64\mathbb{Z}$, and work in the real orbit closure of $64\mathbb{Z}P$. Choose a compact real neighborhood for H around $[h_*]P$ on which the point coordinates, both difference charts, and the six tests are defined and bounded. Such a neighborhood exists: the rational functions restricted to the multiplication curve $H \mapsto (H, [n]H)$ are not identically singular. The difference charts exclude $h = \pm 1$, while every nonzero multiple of the non-torsion P has nonzero abscissa and ordinate. Pass to a smaller compact neighborhood to avoid all remaining zero denominators. Enlarge L with $64 \mid L$ and the real component-group condition so that the progression $h \equiv h_* \pmod{L}$ preserves fixed lower valuation bounds for these tests at $2, 3, 17$. These are precisely the local-continuity and real-orbit steps of the source realization, with fewer point roles and no scalar constraint.

Choose a fixed D_0 with $D_0^2\theta/a_r \in \mathbb{Z}$ for all contacts and both scales, set $v = D_0^2 w^2$, and seek $\lambda = u/v$. Apply the retained prime-pattern theorem to the distinct forms $u - (\theta/a_r)v$, after repetitions have been removed. All bad primes and previously fixed moduli are excluded before this application. The theorem permits arbitrarily large divisibility of w, a sufficiently large prescribed lower bound on u/w^2, and the conjugate reduction-order conditions.

As in [3, 4], the annihilating order moduli U_0, V_0 for the two conjugate families satisfy

$$\gcd(U_0, V_0) = 1, \qquad \gcd(U_0, L) = 16, \qquad \gcd(V_0, L) = 1.$$

Thus the congruences $h \equiv 0 \pmod{U_0}$, $h \equiv 1 \pmod{V_0}$, $h \equiv h_* \pmod{L}$ are compatible, since $16 \mid h_*$. Choose h in that progression while retaining the real neighborhood and avoiding the finitely many excluded multiples. At one place above each contact prime, H, U are formal; at the conjugate place, $H - P, U - A$ are formal. Both exact logarithmic ratios are n. The

constant approximation has precision four, so all six tests have their assigned order one, including the target $n = \beta$.

The inherited detector preparation now applies without alteration. For clarity, its main ingredients are the following. The identity

$$R_r = \frac{\prod_{\theta\in\Theta}(u - (\theta/a_r)v)}{v^5(a_r\lambda - 1)^6}$$

has a rational-square denominator since v is a square. At each chosen contact prime $v_p(R_r) = 1$; the successful conjugate place gives all inverse norm-factor residues equal to one, and the trace of W_r^{-1} is a unit with residue one or two. The resulting b_r has square reduced denominator, and the prescribed congruences give its local squareclasses at $2, 3, 17$. The fixed lower bounds on the tests and sufficient divisibility of w give the depths in (1). On the compact real set, uniformly,

$$R_r \sim a_r^{-6}\lambda^{-1}, \qquad W_r = 1 + O(\lambda^{-3}), \qquad b_r \sim (4a_r^6\lambda)^{-1}.$$

Thus $0 < b_r < 17$ for the sufficiently large lower bound on λ fixed before prime selection. All detector guards are nonzero. These calculations are the source comparison-pole preparation proof [4]; they depend on the finite list of tests and their orders, not on a Lucas or scalar test.

Each prepared input has an independent pole certificate with $pqs \neq 0$ by the retained pole-realization hypotheses. Since $p \neq 0$ excludes $e = 1$, the substitution $\tau = p/(e - 1)$ gives the single chart used above and its nonzero denominator. Thus both nonzero quadratic-root conditions are realized. The direct point charts cover A, H, U. Every rational input $X \in \mathbb{Z}$ therefore has all required witnesses, for every fixed odd $B \geq 3$.

7 The polynomial, its bounds, and verification

The retained coordinates are exactly

Coordinates	Number
Input X	1
Three point pairs for A, H, U	6
Comparison slope λ	1
Two independent pole pairs (τ_r, s_r)	4
Retained before finite-root elimination	12

There are three monic point quartics and two monic pole quadratics. The point algebra has rank $4^3 = 64$, and the full algebra has rank $4^3 2^2 = 256$. Its complete guard is gq_1q_2 with g in the point algebra, and each quadratic radicand also belongs to that algebra. There are no further residual equations. The separating-element theorem [7] therefore replaces all five roots and all nonzero guards by one rational coordinate, without changing the input projection. This gives thirteen total coordinates. Its determinant formulas and its 64-factor product, applied to the supplied integer coefficient circuit, define F_B completely.

The native generator weights are

$$6, 6, 6, 50912, 76641.$$

The full guard and its point factor have weights 156672 and 29119. In the notation of [7, Theorem 4.1], the multiplication-element weight is at most $W_1 r + W_0$, where

$$W_1 = 10200912, \qquad W_0 = 75650993.$$

Thus

$$\deg F_B \le 256(10200912 \cdot 64^2 + (75650993 + 1) \cdot 127) = 13155996618240 < 2^{44}.$$

The input shift is a named coefficient parameter of weight zero in the coefficient graph. No operation or branch depends on its numerical value. Consequently these weight inequalities, and hence the degree bound, hold symbolically for every fixed B.

The directory `computation/few_variables/conditional13` contains the rational-source builder, its integer coefficient specification, and independent source and bound certificates. The source has 1731 rational-circuit nodes and its integer coefficient graph has 748 nodes. The independent conversion auditor checks every affine substitution, the three point quartics and their automatic nonzero factors, both pole-square clearings, and all 46 atomic guard factors by exact expression identities. The separate finite-algebra checker verifies coefficient dependencies, companion-column inequalities and all 64 degree cases without importing the builder.

For the algebraic fixture $B = 3$ it certifies

$$\log_2 H(F_3) \le 31315024824505 < 2^{45}.$$

This is not a height bound for a member certified to define $\mathbb{Z}$. The proof provides an ineffective sufficient shift, whereas the coefficient algorithms construct and bound each specified member. The exact finite checks establish their recorded algebraic assertions; they do not verify *abc* or the inherited arithmetic premises.

References

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